

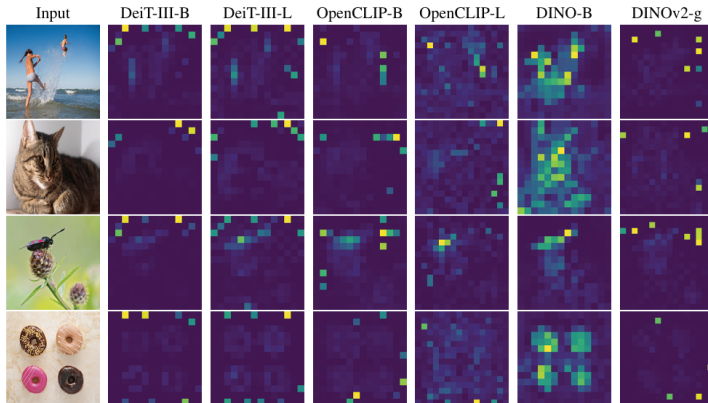
Vision Transformers Need Registers

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ADL Presentation-I

Motivation

- Vision Transformers (ViTs) are powerful for learning image representations.
- However, modern ViTs show artifacts in their feature and attention maps.
- These artifacts appear as high-norm tokens in background regions.



Observation

- DINO models produce clean, interpretable attention maps.
- Other ViTs (DeiT-III, OpenCLIP, DINOv2) show outlier tokens with $10\times$ higher norms.
- These outliers correspond to background patches with little information.

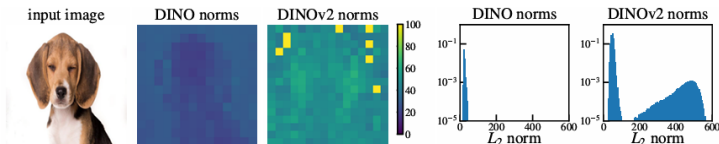
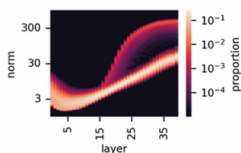


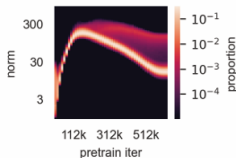
Figure: Attention maps showing artifacts in different ViTs (source: paper Fig. 2).

Artifacts in the Local Features of DINOv2

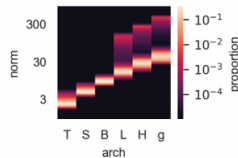
- Artifacts = $10\times$ high-norm tokens (2% of total).
- Appear around mid-layers, sufficient long training and in large models (ViT-L, ViT-g).
- Feature norm distribution is bimodal — few tokens have extreme values.



(a) Norms along layers.



(b) Norms along iterations.

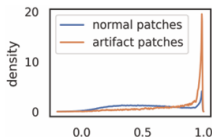


(c) Norms across model size.

Figure: Illustration of several properties of outlier tokens in the 40-layer DINOv2 ViT-g model.

Quantitative Analysis of Tokens

- High-norm tokens appear where patch information is redundant - Verification by **cosine similarity**.
- High-norm tokens hold little local information.
 - Position prediction
 - Pixel reconstruction
- Linear models trained on patch embeddings.
- High-norm tokens → poor at predicting position or reconstructing pixels.
- Artifacts hold **global information** - Artifacts show high accuracy while predicting the image class.



(a) Cosine similarity to neighbors.

	position prediction		reconstruction
	top-1 acc	avg. distance ↓	L2 error ↓
normal	41.7	0.79	18.38
outlier	22.8	5.09	25.23

(b) Linear probing for local information.

Hypothesis

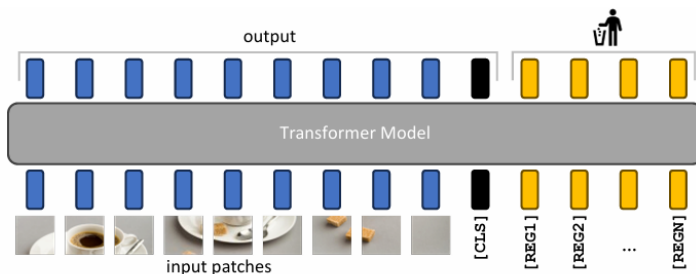
Main Hypothesis

Large ViT models learn to recognize patches containing little useful information, and recycle the corresponding tokens to aggregate global image information while discarding spatial information.

- Explains appearance of high-norm tokens.
- Suggests that ViTs internally perform computations within a restricted subset of tokens.
- Behaviour not bad, but it happens inside the patch tokens is undesirable.

Proposed Solution: Register Tokens

- Add learnable, image-independent tokens after patch embedding layer (called registers).
- Registers act as dedicated places for global computations.
- This prevents patch tokens from losing local information.



Experiments

- Models trained: DeiT-III (supervised), OpenCLIP (text-supervised), DINOv2 (self-supervised).
- Dataset: ImageNet-22k, Shutterstock dataset containing image-caption pairs.

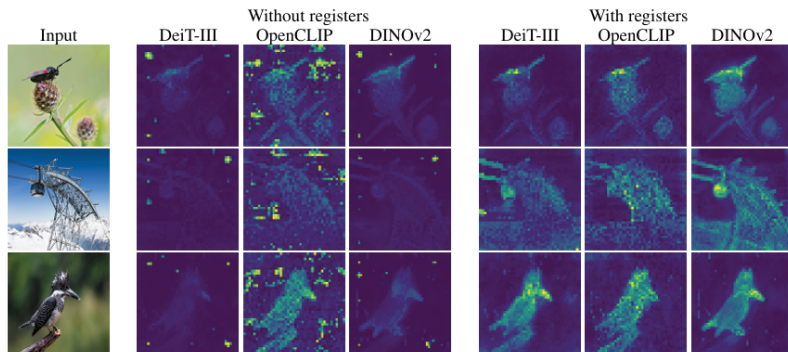


Figure: Register tokens remove norm outliers.

Experiments

- Performance regression - The use of register tokens does not affect the representation quality of those features
- Linear probing on ImageNet classification, ADE20k Segmentation, and NYUd monocular depth estimation.

	ImageNet Top-1	ADE20k mIoU	NYUd rmse ↓
DeiT-III	84.7	38.9	0.511
DeiT-III+reg	84.7	39.1	0.512
OpenCLIP	78.2	26.6	0.702
OpenCLIP+reg	78.1	26.7	0.661
DINOv2	84.3	46.6	0.378
DINOv2+reg	84.8	47.9	0.366

(a) Linear evaluation with frozen features.

	ImageNet Top-1
OpenCLIP	59.9
OpenCLIP+reg	60.1

(b) Zero-shot classification.

Figure: Register tokens remove norm outliers.

Experiments

- Number of register tokens - They trained DINOv2 ViT-L/14 models with 0, 1, 2, 4, 8 or 16 registers.
- Artifacts disappear when adding at least one register.
- There seems to be an optimal number of registers for dense tasks, and adding one brings most of the benefit.

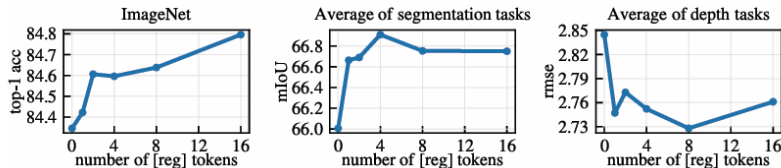
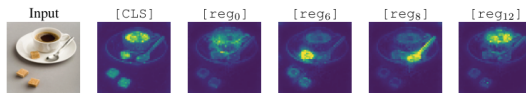


Figure: Performance w.r.t. no. of registers

Object Discovery and Qualitative Evaluation of Registers

	VOC 2007	VOC 2012	COCO 20k
DeiT-III	11.7	13.1	10.7
DeiT-III+reg	27.1	32.7	25.1
OpenCLIP	38.8	44.3	31.0
OpenCLIP+reg	37.1	42.0	27.9
DINOv2	35.3	40.2	26.9
DINOv2+reg	55.4	60.0	42.0



- Some selected registers exhibit interesting attention patterns, attending to the different objects in the scene.

Conclusion

- Artifact tokens can be detected via outlier norms and removed by appending extra register tokens.
- This fix improves dense prediction and object discovery, while also removing artifacts in supervised models like DeiT-III and OpenCLIP.
- Attention maps become smoother and more interpretable, confirming the generality and effectiveness of the approach.